Heart Disease Prediction: End-to-End MLOps Pipeline

Course: Machine Learning Operations (AIMLCZG523), Assignment 1
Student: <*your name*> (2025AE05141)
Repository: https://github.com/khuranaj95/heart-disease-mlops
Environment: BITS Prayogshala Linux server, Python 3.12.9, Docker 28, Minikube (Kubernetes v1.32)

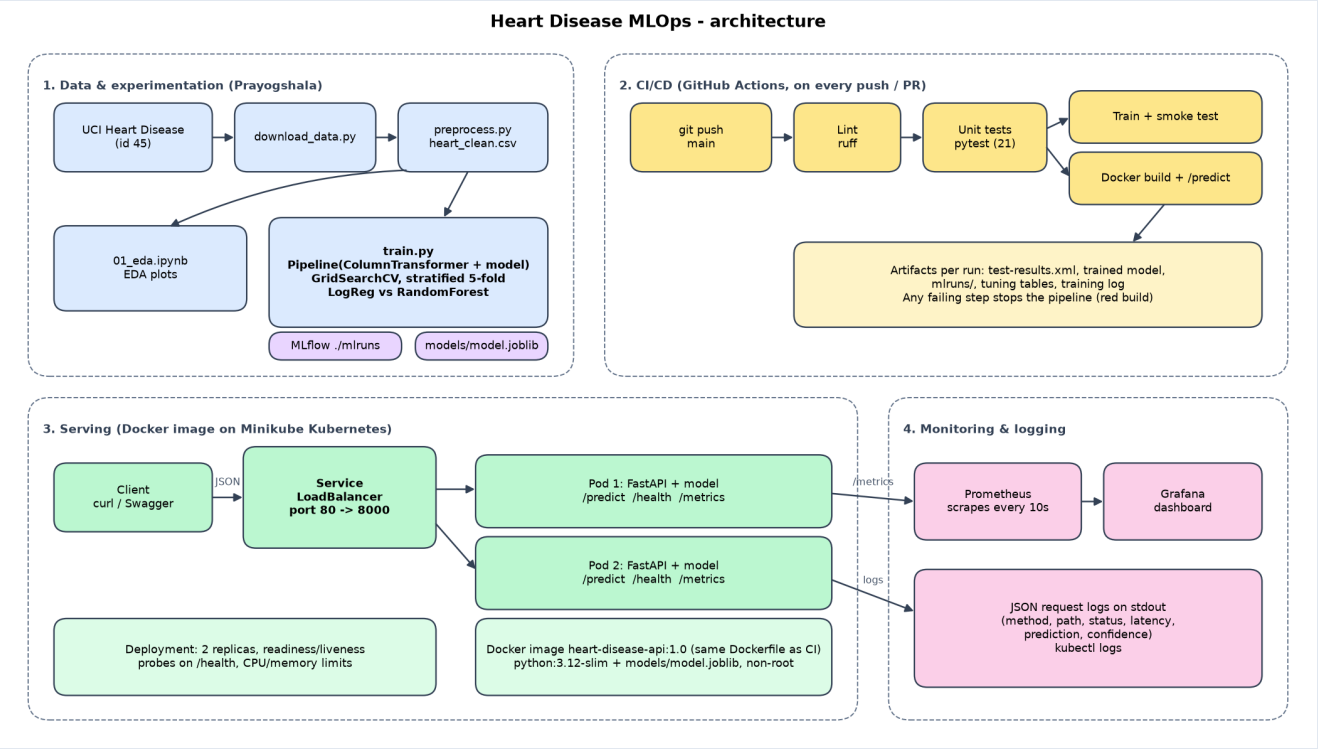
1. Summary

This project builds a classifier that predicts the presence of heart disease from 13 clinical measurements (UCI Heart Disease, Cleveland subset, 303 patients), and takes it all the way to a monitored API running on Kubernetes:

Stage	What was built
Data & EDA	Download script, cleaning script, EDA notebook with 4 figures
Modelling	scikit-learn pipeline, Logistic Regression vs Random Forest, GridSearchCV with stratified 5-fold CV
Tracking	MLflow: 2 parent runs and 16 nested child runs (one per grid setting), metrics, plots, models
Packaging	Full pipeline saved as `model.joblib` and as an MLflow model; pinned `requirements.txt`
CI/CD	GitHub Actions: lint → tests → train, plus Docker build and smoke test
Serving	FastAPI `/predict` in a Docker image (python:3.12-slim, non-root)
Deployment	Minikube: Deployment (2 replicas, probes) + LoadBalancer Service
Monitoring	`/metrics` (prometheus_client), JSON request logs, Prometheus + Grafana dashboard

Final model: Logistic Regression (C = 1.0) with **cross-validated ROC-AUC 0.902** and, on the held-out test set, ROC-AUC 0.966, recall 0.929 and accuracy 0.885.

2. Architecture



The system has four parts. **(1)** Data is downloaded, cleaned and used to train and compare models, with every run logged to MLflow. **(2)** On every push, GitHub Actions lints the code, runs the unit tests, retrains the model and builds/tests the Docker image; any failure stops the pipeline. **(3)** The Docker image (code + trained pipeline) runs as two pods behind a LoadBalancer service on Minikube. **(4)** Prometheus scrapes each pod's `/metrics` endpoint, Grafana visualises it, and every request is logged as a JSON line.

3. Setup and installation

```
git clone https://github.com/khuranaj95/heart-disease-mlops.git
cd heart-disease-mlops
python3.12 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt

python scripts/download_data.py                 # data/raw/heart_raw.csv
python src/preprocess.py                        # data/processed/heart_clean.csv
jupyter nbconvert --to notebook --execute --inplace notebooks/01_eda.ipynb
python -m src.train                             # MLflow runs + models/model.joblib
python -m pytest -v                             # 21 unit tests
python -m src.predict data/sample_request.json  # inference from the command line
```

Docker and Kubernetes:

```
docker build -t heart-disease-api:1.0 .
docker run -p 8000:8000 heart-disease-api:1.0          # http://localhost:8000/docs

minikube start --driver=docker
minikube image load heart-disease-api:1.0
kubectl apply -f k8s/                                  # API deployment + service
kubectl apply -f k8s/monitoring/                       # Prometheus + Grafana
minikube tunnel                                        # gives the LoadBalancer an
external IP
minikube service heart-disease-api --url               # alternative access URL
```

Reproducibility was verified by creating a brand-new virtual environment from `requirements.txt` only and re-running every step: all scripts ran and produced identical metrics.

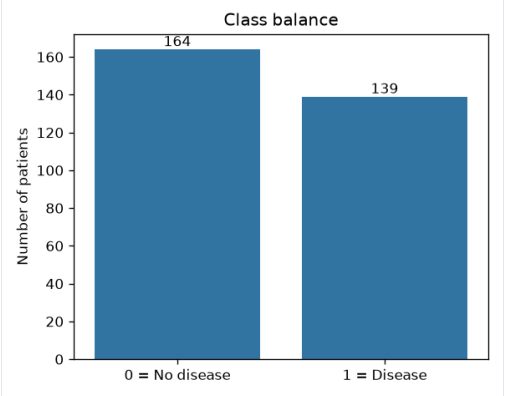
4. Data acquisition and EDA

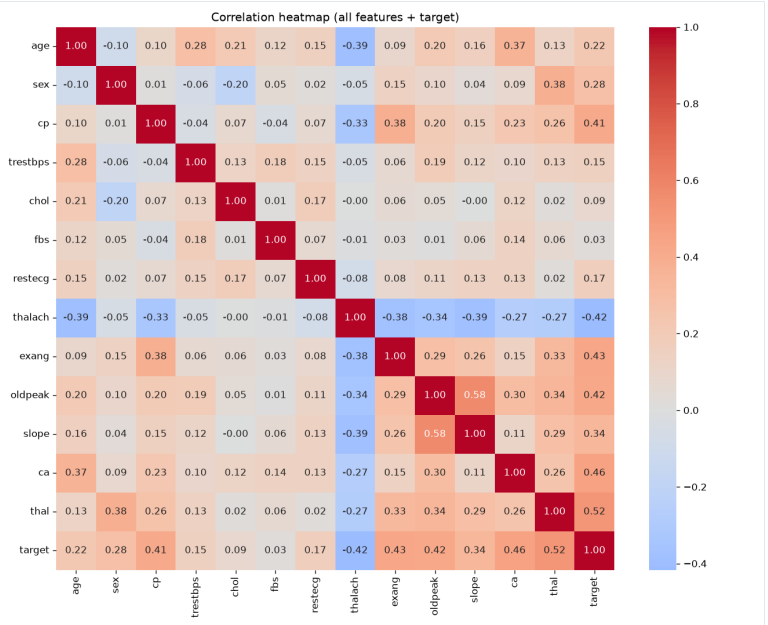
Source. `scripts/download_data.py` fetches dataset id 45 from the UCI repository with `ucimlrepo` and saves it unchanged to `data/raw/heart_raw.csv` (303 rows, 13 features + `num`).

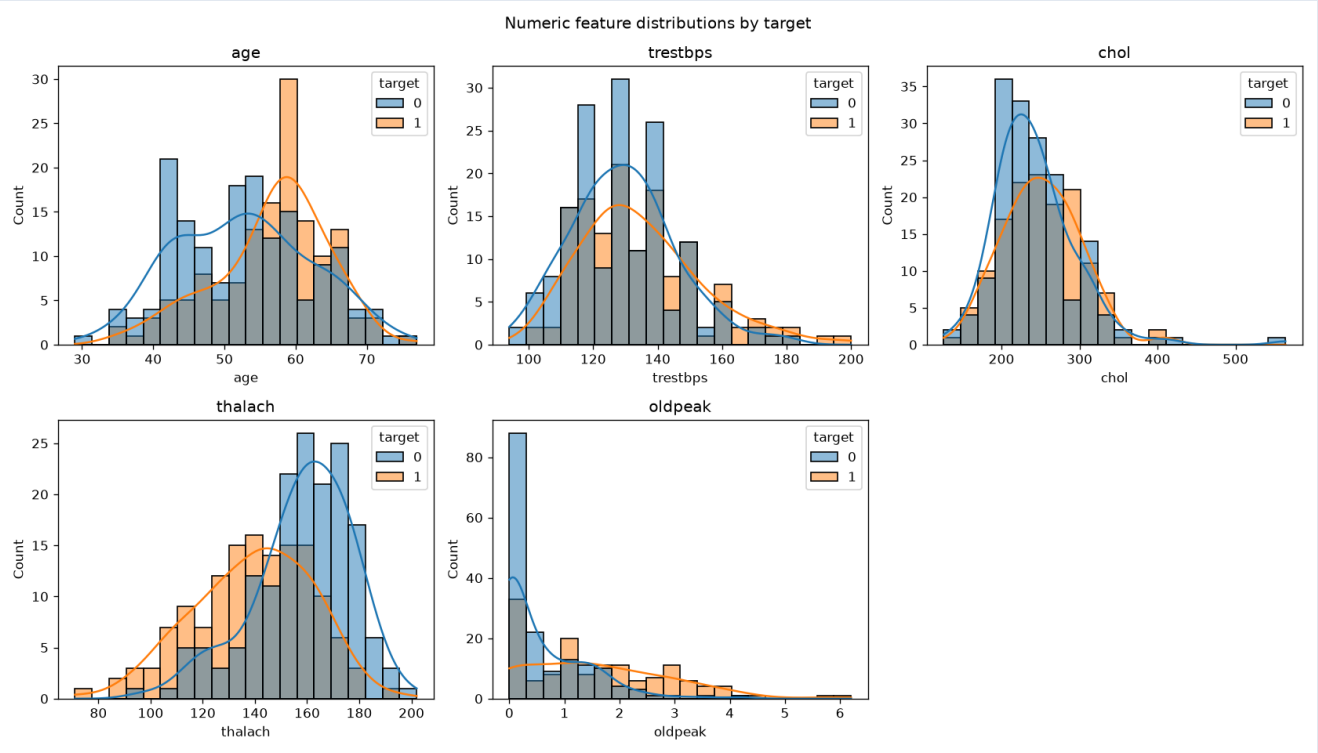
Cleaning (`src/preprocess.py`):

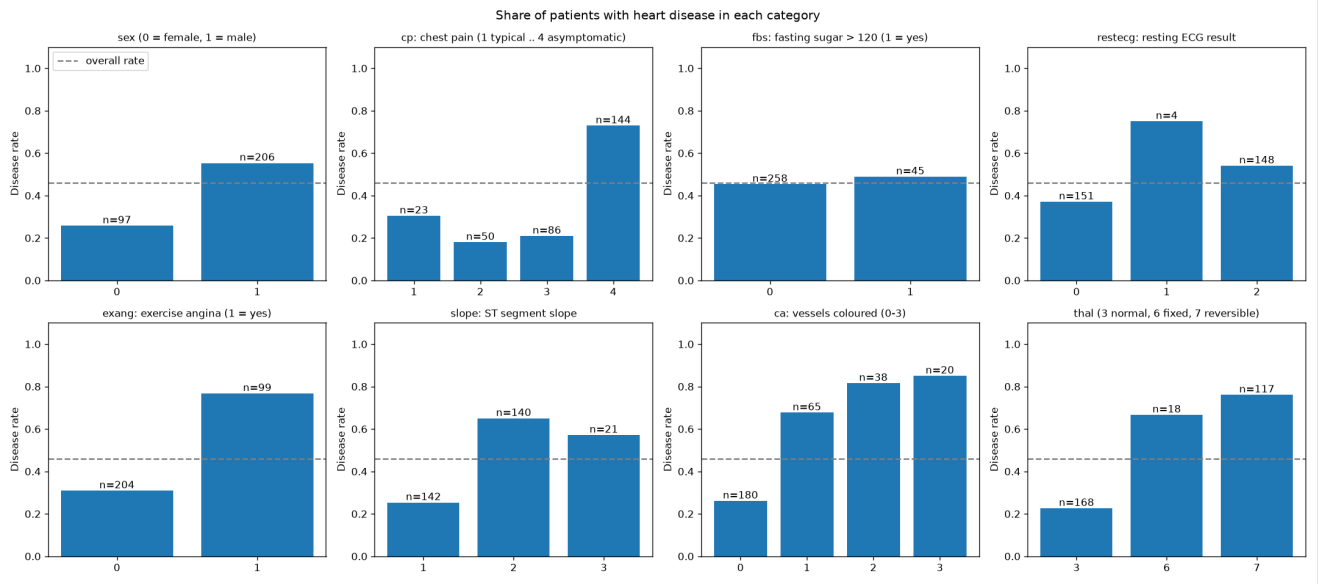
- **Missing values:** only 6 of 303 values are missing (4 in `ca`, 2 in `thal`). Both are categorical, so they are filled with the **mode** rather than the mean (a mean like "0.67 vessels" is not a valid category) and no patients are dropped from an already small dataset. The script also converts the `?` markers used in the original UCI file.
- **Target:** the original `num` (0 = healthy, 1-4 = increasing severity) is converted to binary: 0 → 0, 1-4 → 1.

Class balance. 164 patients without disease vs 139 with disease (54% / 46%): nearly balanced, so accuracy is meaningful, but precision, recall and ROC-AUC are reported too.









Key findings.

- Strongest correlations with the target: `thal` (0.52), `ca` (0.46), `exang` (0.43), `oldpeak` (0.42), `thalach` (-0.42) and `cp` (0.41). `chol` (0.09) and `fbs` (0.03) are weak on their own.
- Patients with **asymptomatic** chest pain (`cp` = 4) have a 73% disease rate versus 18-30% for the "typical" pain types, a counter-intuitive but well-known property of this dataset.
- The disease rate climbs steadily with the number of coloured vessels (`ca`): 26% at 0 up to 85% at 3.
- `restecg` = 1 has only 4 patients: a rare category, which is why the encoder ignores unseen values.
- Features are on very different scales (`chol` ≈ 250, `oldpeak` ≈ 1), so scaling is needed.

5. Feature engineering and model development

Preprocessing (`src/features.py`) is a `ColumnTransformer` inside a scikit-learn `Pipeline` :

Feature group	Columns	Steps	Reason
Numeric	age, trestbps, chol, thalach, oldpeak	median imputer → StandardScaler	median is robust to outliers (chol = 564); scaling matters for Logistic Regression
Categorical	sex, cp, fbs, restecg, exang, slope, ca, thal	most-frequent imputer → OneHotEncoder(handle_unknown="ignore")	codes are labels, not quantities; unseen codes do not crash the API

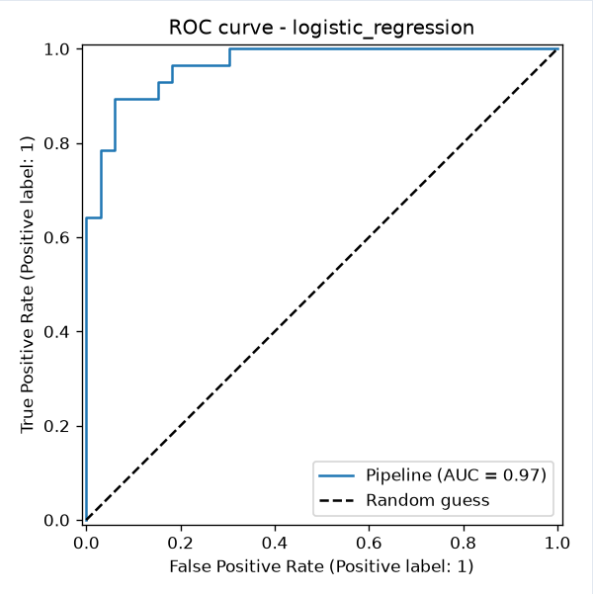
Because preprocessing lives inside the pipeline, it is fitted only on the training folds during cross-validation (no data leakage), and the saved model applies exactly the same transformations at prediction time.

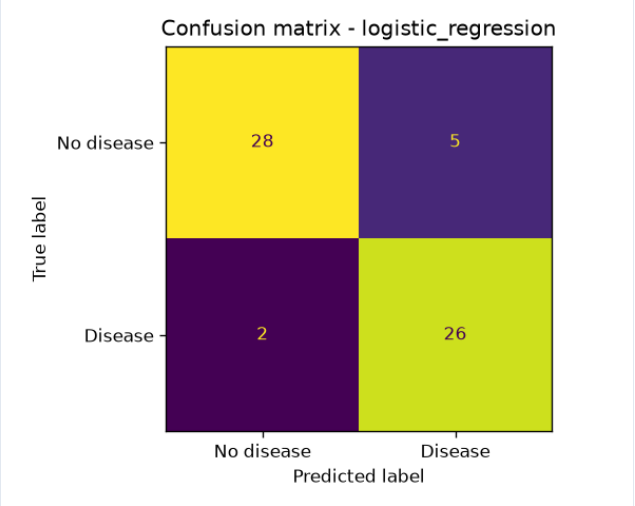
Training procedure (`src/train.py`): stratified 80/20 train/test split (242 / 61 patients, random_state = 42). Each model is tuned with `GridSearchCV` using **stratified 5-fold CV** scored on ROC-AUC, while accuracy, precision and recall are recorded for every fold.

Model	Grid	Settings
Logistic Regression	C ∈ {0.01, 0.1, 1, 10}	4
Random Forest	n_estimators ∈ {100, 300}, max_depth ∈ {None, 5, 10}, min_samples_leaf ∈ {1, 5}	12

Results

Model	Split	Accuracy	Precision	Recall	ROC-AUC
Logistic Regression (C=1.0)	CV	0.847	0.877	0.783	**0.902**
	Test	0.885	0.839	**0.929**	**0.966**
Random Forest (300 trees, leaf=5)	CV	0.810	0.810	0.774	0.902
	Test	0.852	0.806	0.893	0.948





Model selection: Logistic Regression. The model is chosen on **cross-validated** ROC-AUC, never on the test set, which stays untouched until the end. The two models are effectively tied (0.9025 vs 0.9023), so the decision used the secondary criteria:

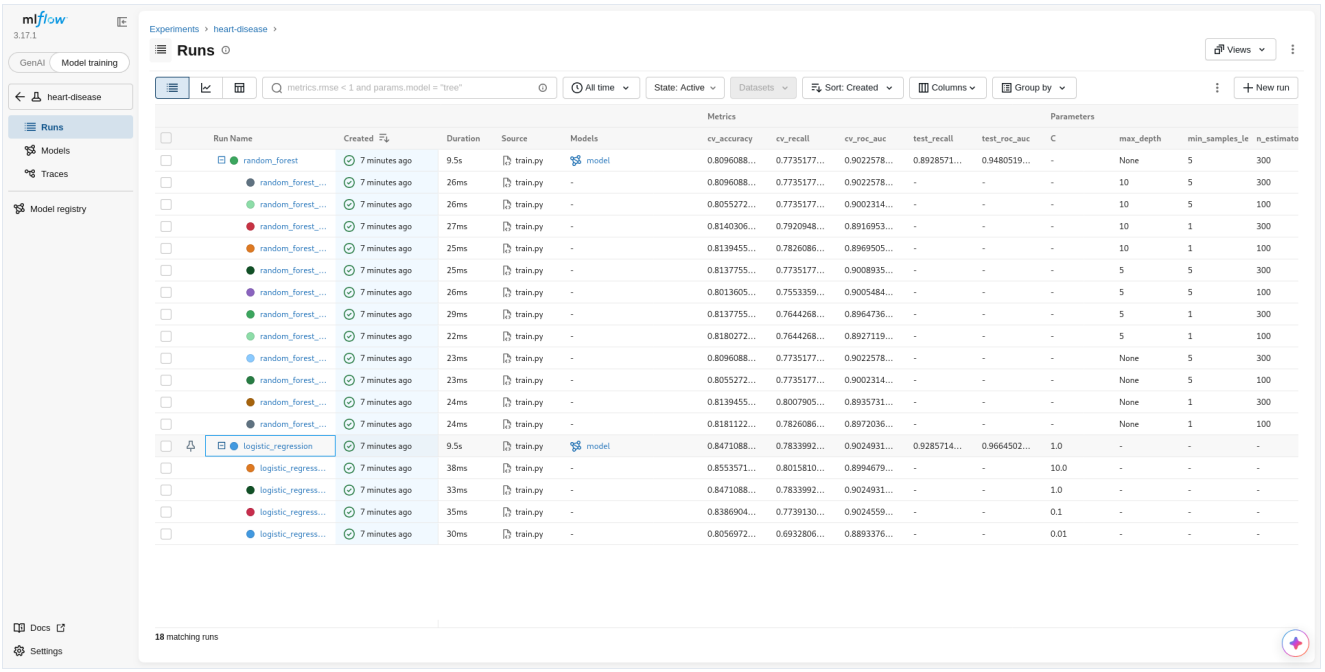
1. **Recall:** a missed sick patient is worse than a false alarm; LR has higher recall in CV (0.783 vs 0.774) and finds 26 of the 28 sick test patients.
2. **Stability:** LR's fold-to-fold ROC-AUC spread is ±0.014 versus ±0.033 for the forest.
3. **Simplicity:** one interpretable weight per feature, which suits a 303-patient dataset.

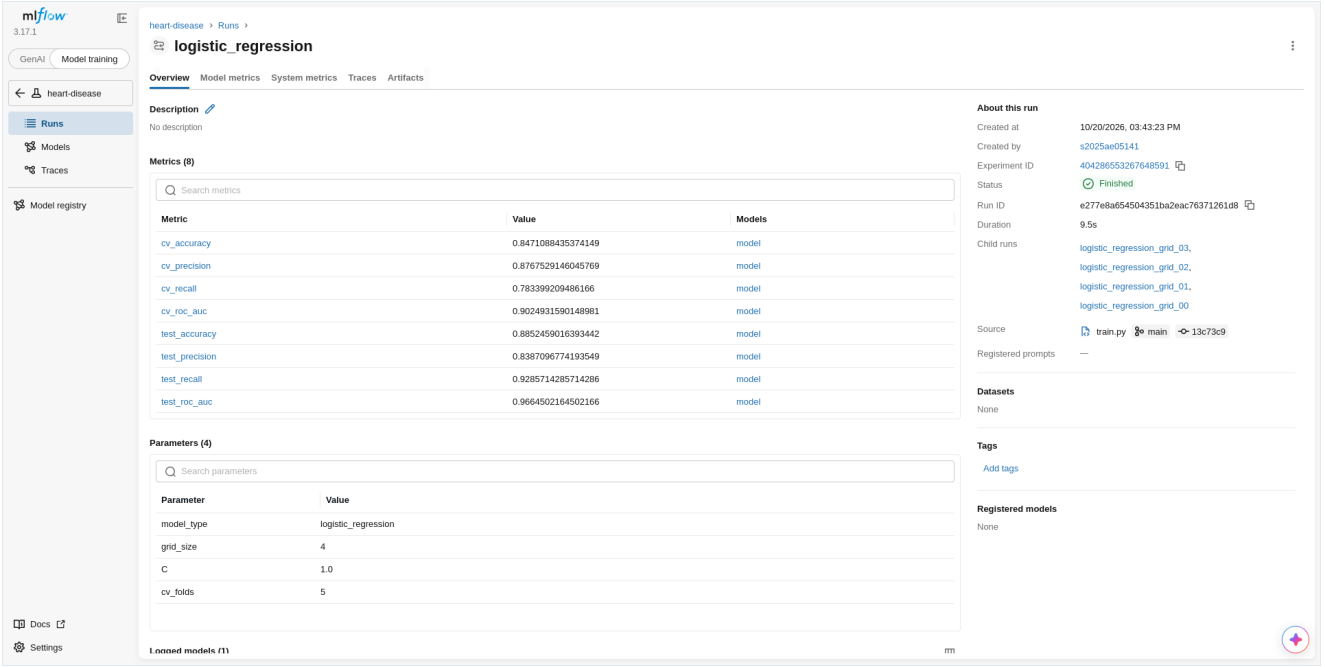
Tuning observations: very strong regularisation (C = 0.01) costs a lot of recall (0.69), and C between 0.1 and 10 forms a flat plateau. For the forest, every top setting uses `min_samples_leaf = 5`, which limits overfitting on the small dataset. Test scores are higher than CV scores because 61 patients is a small, noisy test set; ~0.90 ROC-AUC is the realistic expectation. Full tables: `report/model_selection.md` and `report/tuning/`.

6. Experiment tracking (MLflow)

All training is logged to the MLflow experiment **`heart-disease`** (local `./mlruns` store):

- **One parent run per model** with the best parameters, 4 CV metrics and 4 test metrics, the ROC curve and confusion matrix PNGs, the full grid table (CSV), and the fitted pipeline logged as an MLflow model with an input signature and example.
- **One nested child run per grid setting** (16 in total) with its parameters, CV metrics and rank, so the entire tuning process is visible, not only the winner.





Further screenshots in `screenshots/mlflow/` : side-by-side comparison, ROC and confusion matrix artifacts, and the tuning CSV artifact.

7. Model packaging and reproducibility

- **Saved model:** `models/model.joblib` holds the *whole* pipeline (imputers, scaler, encoder and classifier), so raw patient data goes in and a prediction comes out. The same pipeline is logged as an MLflow model, and `models/metadata.json` records the feature list, metrics, MLflow run ID and exact library versions.
- **Inference:** `src/predict.py` loads the model once and returns `prediction` , `label` , `confidence` and `probability_disease` ; it is reused by the API.
- **Environment:** `requirements.txt` pins every library to an exact version; `requirements-serve.txt` pins the 7 libraries the API image needs (same versions).
- **Determinism:** fixed `random_state = 42` for the split, CV folds and Random Forest.

8. CI/CD pipeline and automated testing

Unit tests (`tests/` , 21 tests, pytest):

File	What it checks
`test_preprocess.py`	? /NaN handling, mode filling, binary target, integer types, input not modified
`test_features.py`	feature lists, output shape, scaling (mean 0 / std 1), missing and unseen values
`test_predict.py`	output format, label/probability consistency, known patients, missing feature
`test_train.py`	model configuration, metric ranges, **quality gate: CV ROC-AUC ≥ 0.85**
`test_api.py`	`/health` , `/predict` response, 422 on invalid or missing input, `/metrics` counters

GitHub Actions (`.github/workflows/ci.yml`) runs on every push and pull request to `main` :

1. **Lint:** `ruff check` and `ruff format --check` .
2. **Unit tests:** `pytest -v` , JUnit report uploaded as an artifact.
3. **Train model:** clean data, train, smoke-test the model; uploads the model, `mlruns/` , tuning tables and training log as artifacts.
4. **Docker:** build the image, start it, wait for `/health` , call `/predict` and check the answer.

Jobs depend on each other (`needs:`), every step runs under `bash -eo pipefail` , and the quality gate turns a model regression into a failing test, so **any code, test or model error produces a red build with the failing step's log**.

> *[Add: screenshot of a green workflow run (all 4 jobs) and of the run's artifacts list.]*

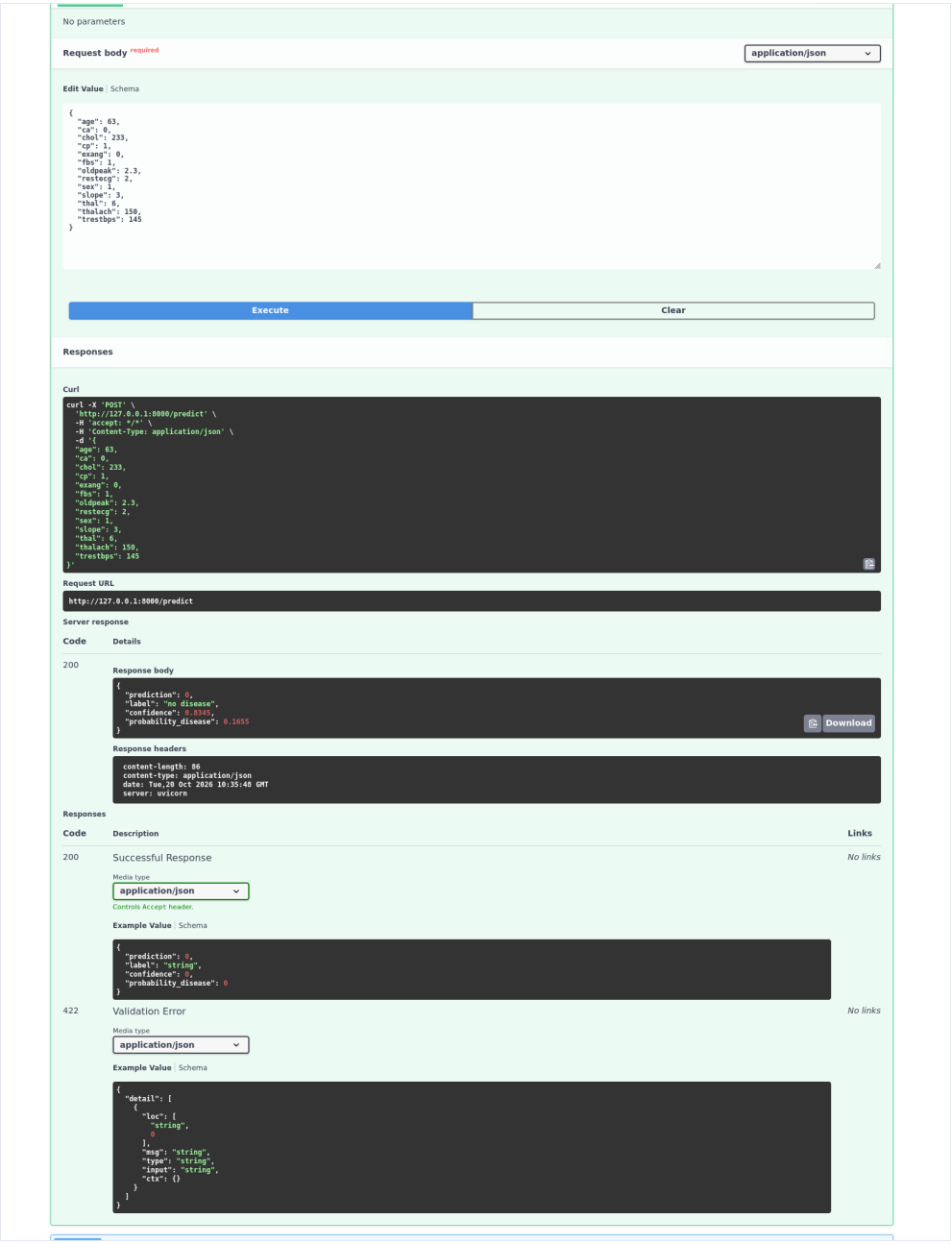
9. Model containerisation

`src/api.py` is a FastAPI service:

- `POST /predict` takes the 13 features as JSON. Pydantic validates types and ranges (e.g. `age` 1-120, `thal` ∈ {3, 6, 7}) and returns HTTP 422 with a clear message for bad input.
- It returns `{"prediction": 1, "label": "disease", "confidence": 0.9856, "probability_disease": 0.9856}` .
- `GET /health` is used by Docker and Kubernetes probes, and `GET /metrics` by Prometheus.

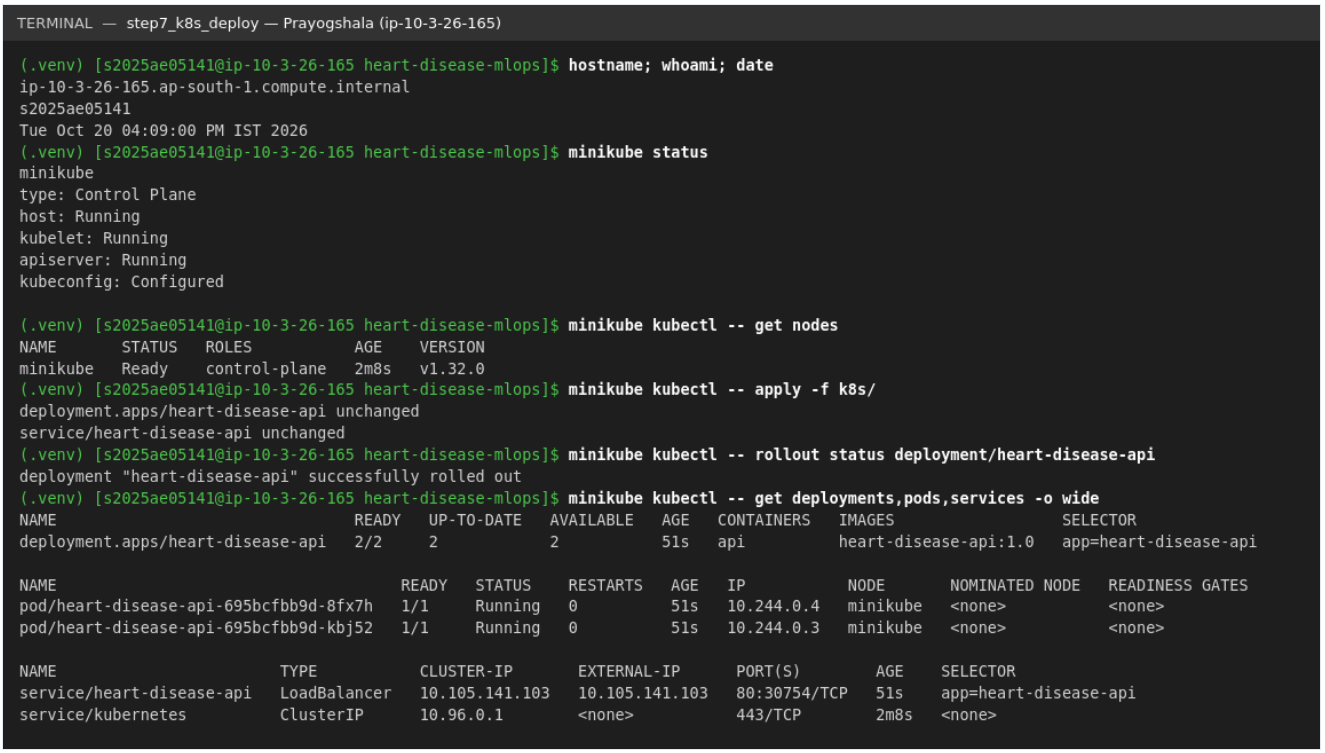
The `Dockerfile` uses `python:3.12-slim` , installs dependencies in a cached layer, copies only `src/` and `models/` , runs as a non-root user and defines a `HEALTHCHECK` . The image is 465 MB.





10. Production deployment (Minikube)

`k8s/deployment.yaml` runs **2 replicas** with CPU/memory requests and limits, a **readiness probe** (no traffic until the model is loaded) and a **liveness probe** (restart if `/health` stops answering). `k8s/service.yaml` exposes them through a **LoadBalancer** service on port 80; `minikube tunnel` assigns its external IP (`10.105.141.103`).



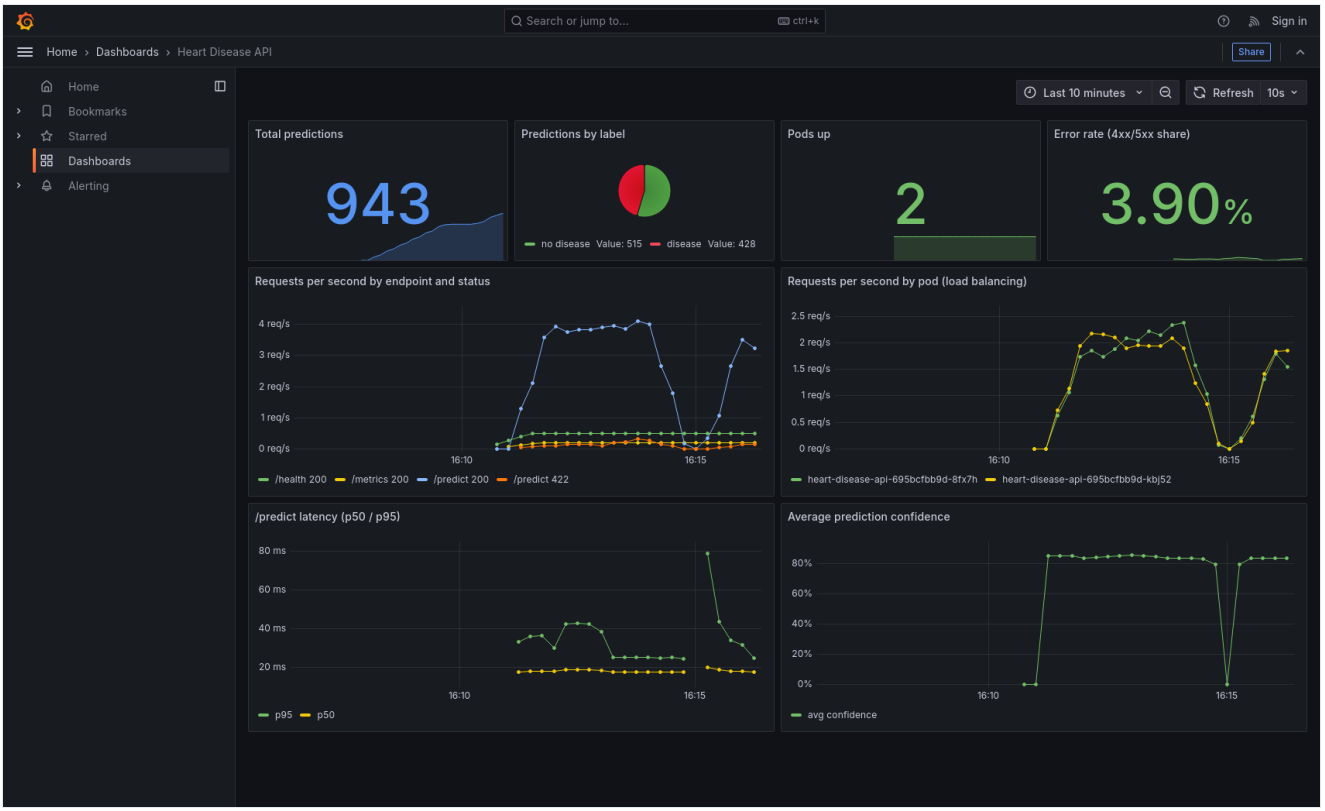


Endpoint verification: `/health` and `/predict` answer through both the LoadBalancer IP and the `minikube service` URL, and 20 consecutive requests were split evenly across the two pods (11 / 11 including earlier calls).

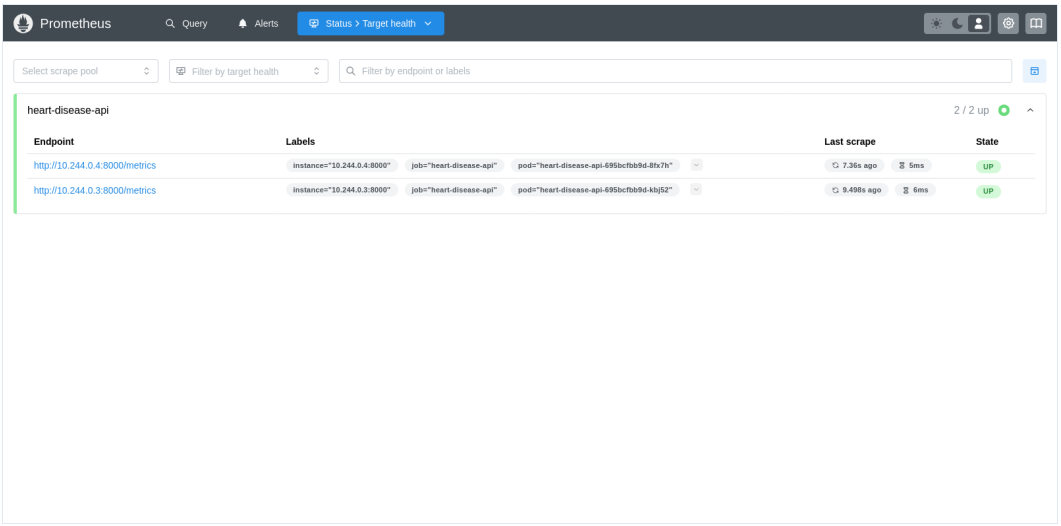
Access instructions (local testing): after `kubectl apply -f k8s/`, run `minikube service heart-disease-api --url`, then `POST <url>/predict` with JSON as in `data/sample_request.json`; interactive docs are at `<url>/docs`.

11. Monitoring and logging

- **Request logging:** middleware writes one JSON line per request (method, path, status, latency, client) plus one per prediction (label, confidence). The patient's health data is deliberately not logged. Logs go to stdout, so `docker logs` / `kubectl logs` collect them.
- **Metrics** (`prometheus_client`): `api_requests_total{method,endpoint,status}`, `api_request_duration_seconds` (histogram), `model_predictions_total{label}` and `model_prediction_confidence` (histogram).
- **Prometheus** discovers the API pods through their `prometheus.io/scrape` annotations and scrapes them every 10 s; **Grafana** comes provisioned with the data source and an 8-panel dashboard.



The dashboard was captured after sending ~950 requests (about 5% deliberately invalid): it shows the prediction split, both pods up, the 3.9% error rate from the invalid requests, traffic spread evenly across the two pods, p50/p95 latency of ~18/25-40 ms, and an average confidence of ~85%.



12. Conclusion and possible improvements

The pipeline goes from a raw UCI download to a tested, versioned, containerised and monitored model service, and every stage runs from a clean environment. Possible next steps:

- Use a remote MLflow tracking server and model registry instead of the local folder store.
- Push the image to a registry from CI and deploy automatically (continuous delivery).

- Add Prometheus alert rules (e.g. error rate > 5%, p95 latency > 200 ms) and data-drift monitoring of the input feature distributions.
- Collect more data: 303 patients limit how far tuning and model choice can be trusted.

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